

Problem Set 3

Date 28.04.2026
Discussion 4. & 5.05.2026

Problem 1 – Orbits

1. What is the distance above Earth's surface of a satellite on a geostationary orbit (i.e., the satellite is above the equator at fixed longitude)? Can a satellite be permanently placed above Munich?
2. Consider a satellite on a highly elliptical orbit around Earth with $r_p = 500$ km and $r_a = 10,000$ km. How can the satellite be brought into a circular orbit? How could the satellite be brought into a geosynchronous orbit (see above)?
3. An advantageous trajectory to Mars is from Earth at its perihelion to Mars at its aphelion. What kind of trajectory does this correspond to, and how long does the journey take?

Problem 2 – Sphericity of Celestial Objects

1. Determine the pressure at the Earth's center! For simplicity, assume a constant density $\rho(r) = \rho_0 = 5.5 \text{ g/cm}^3$ and show that in this case

$$P(r=0) = \rho_0 g_0 \frac{R_{\text{Earth}}}{2} \quad (1)$$

with the gravitational acceleration at the Earth's surface g_0 . What is the value obtained for $P(r=0)$?

2. Establish an expression for the pressure as a function of depth h . Consider the limiting case of small depths, where $g(R-h) \approx g(R) \equiv g_0$ remains valid.
3. Rocks remain stable up to a critical pressure of $P_{\text{crit}} \approx 2 \times 10^8$ Pa. Compute the maximal height of a mountain made of rock with a mean density of 2.5 g/cm^3 . How does the situation change on Mars?
4. How large does a celestial object need to be to become spherical? Compare your finding with objects of the solar system!

Problem 3 – Atmospheres

1. Compute the equilibrium temperature on Mars, assuming an albedo of $A = 0.15$ (for comparison, Earth's albedo is $A = 0.37$)!
2. Compute the escape velocity for Mars and for Earth! Compare these velocities with the thermal velocities of N_2 and CO_2 , and interpret your results!

Problem 4 – Tidal Forces

1. Compute the Roche limit for the Earth–Moon system, assuming $\rho_{\text{Earth}} = 5.5 \text{ g/cm}^3$ und $\rho_{\text{Moon}} = 3.3 \text{ g/cm}^3$. Present your result in units of the Earth's radius! What do you conclude?
2. Which combinations of distance, densities, and planetary radius lead to a large Roche limit?
3. Now consider Saturn, and compute its Roche limit for objects with a density of $\rho = 1.2 \text{ g/cm}^3$. Saturn's density is 687 kg/m^3 and its radius is 6×10^7 m. Where in Saturn's system is the Roche limit? What does this imply?
4. Why are spacecraft not tidally disrupted when they pass near the giant planets?